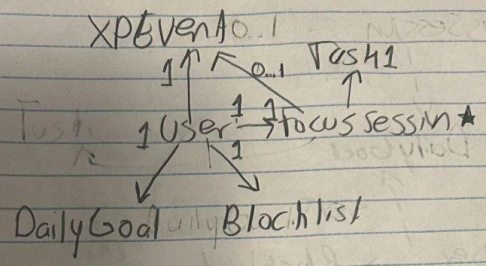
Pratheek Thummdapalli

Class Diagram



User
- -userId: Int
- -username: String
- +startFocusSession(task: Task): void
- +setDailyGoal(target: int): void

Task
- -taskId: int
- -title: String
- +completeTask(): void

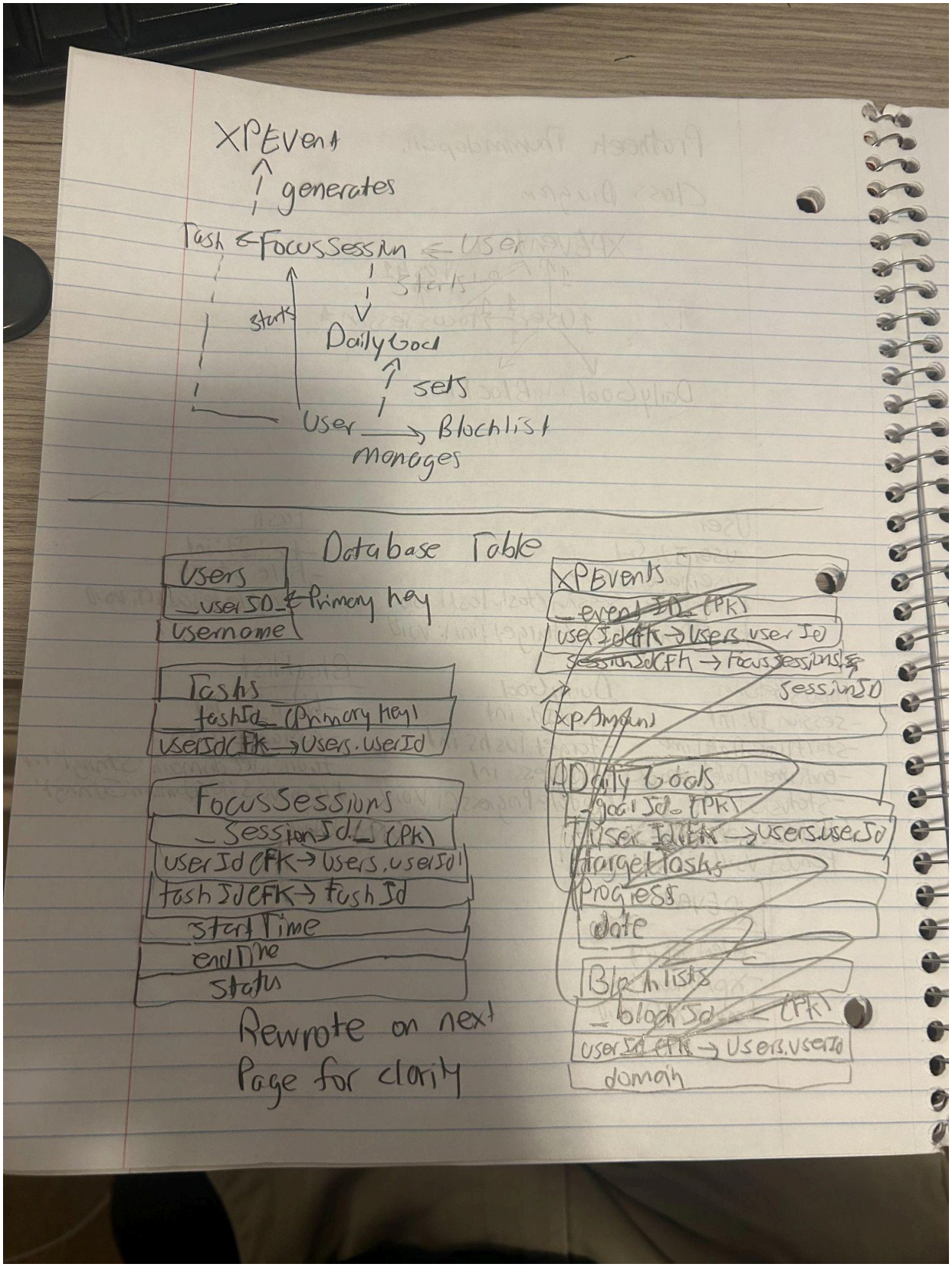
FocusSession
- -sessionId: int
- -startTime: DateTime
- -endTime: DateTime
- -Status: String
- +start(): Void
- +end(): void

DailyGoal
- -goalId: int
- -target Tasks: int
- -progress: int
- +updateProgress(): void

BlockList
- -blockId: int
- -domain: String
- +addSite(domain: String): void
- +removeSite(domain: String): void

XPEvent
- -eventId: int
- -xpAmount: int
- +award XP(): void

My Class Diagram



My ER Diagram and half of my database Table

XPEvents
eventID (Primary key)
userId (Fk → Users.userId
sessionId (Fk → FocusSessions. ← sessionID
xpAmount

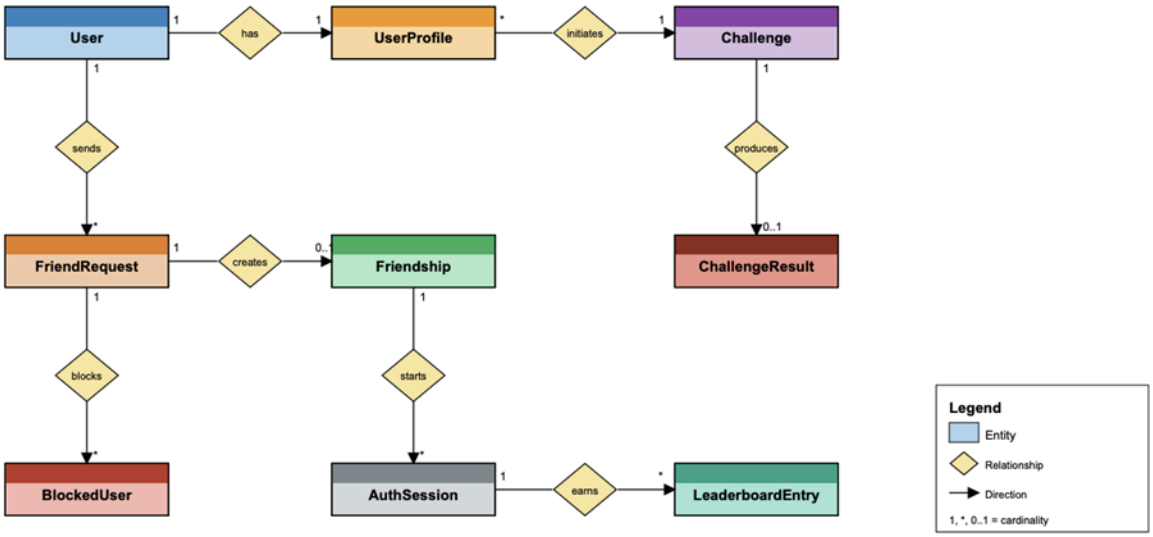
Daily Goals
goalID (Primary key)
user Id (Fk → Users.userId)
targetTasks
Progress
date

Blocklists
blockId (Pk)
userId (Fk → Users.userId
domain

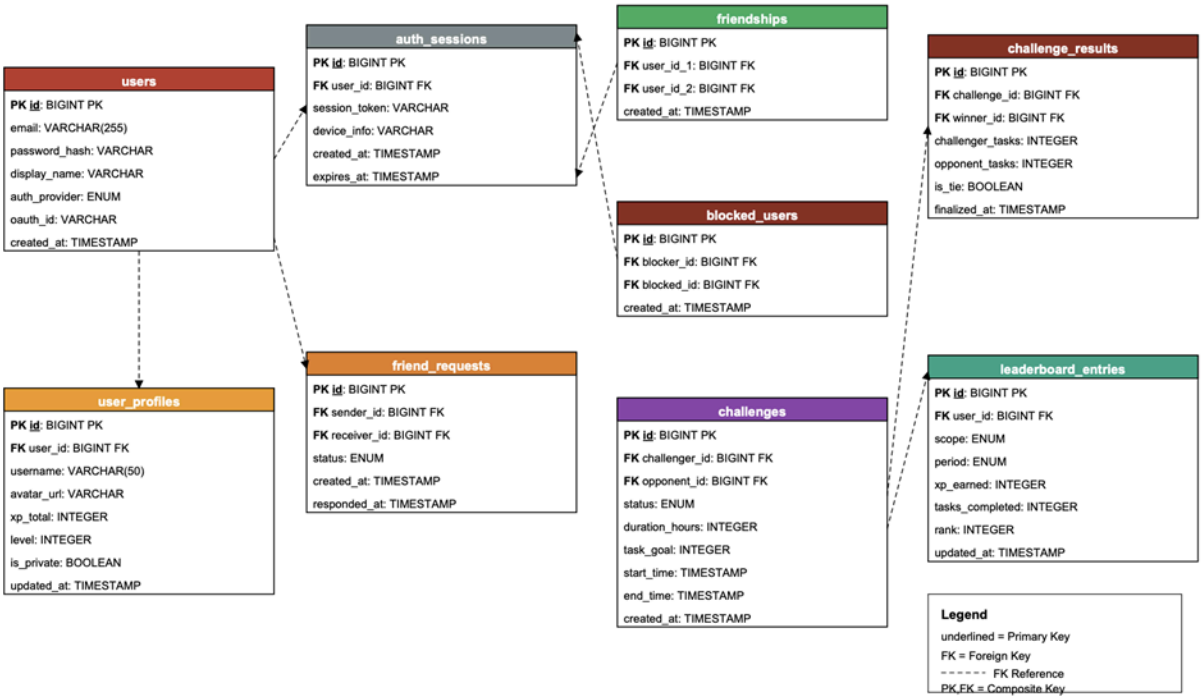
The other written half of the Database Table

Satwik Valluri:

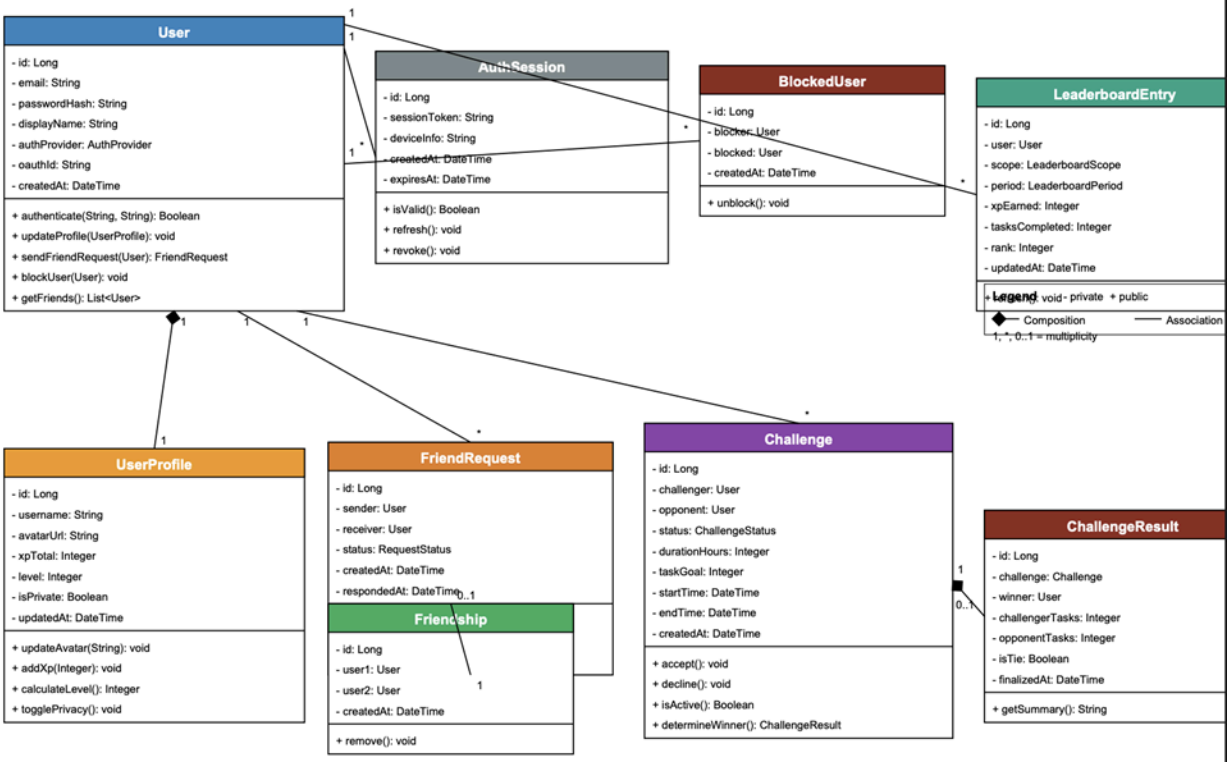
Social & Gamification System - ER Diagram



Social & Gamification System - Database Tables



Social & Gamification System - Class Diagram

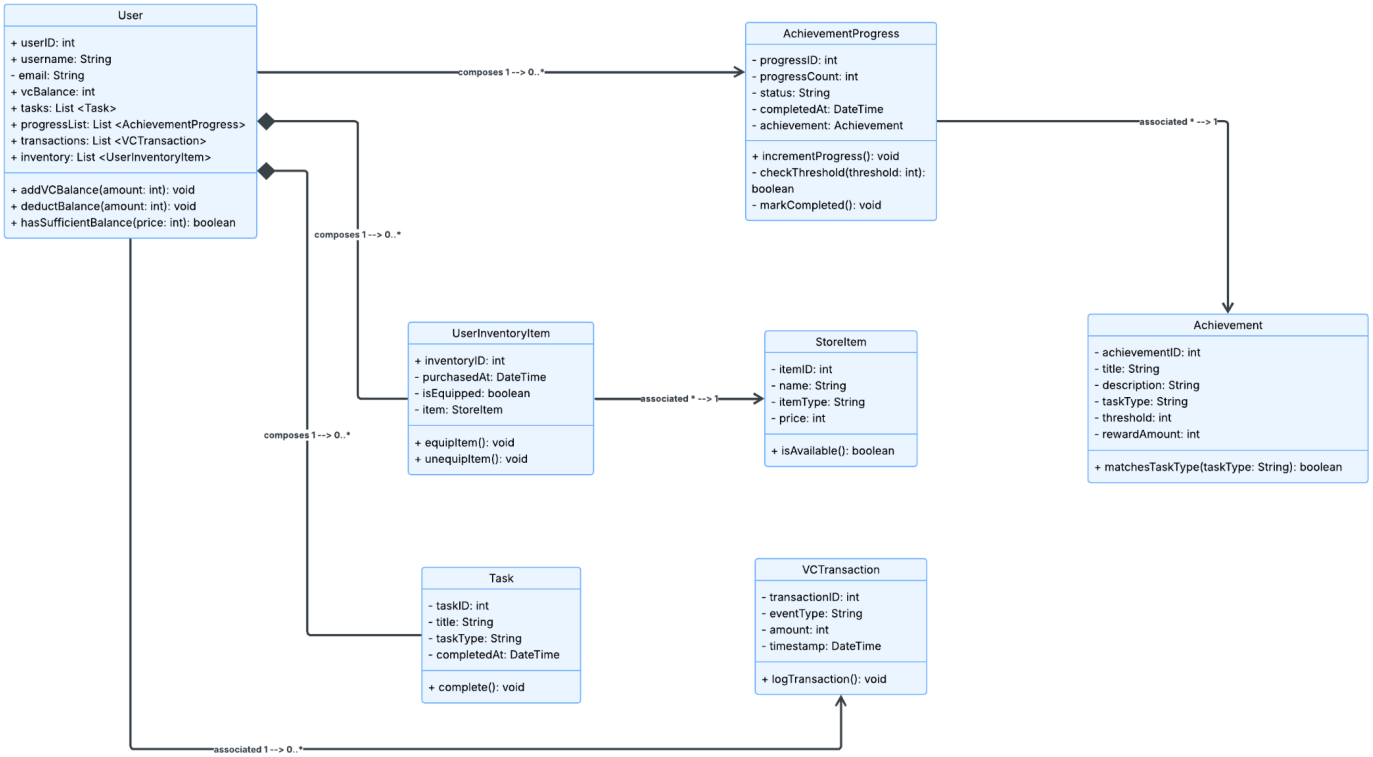


Social & Gamification System - Class Diagram (continued)

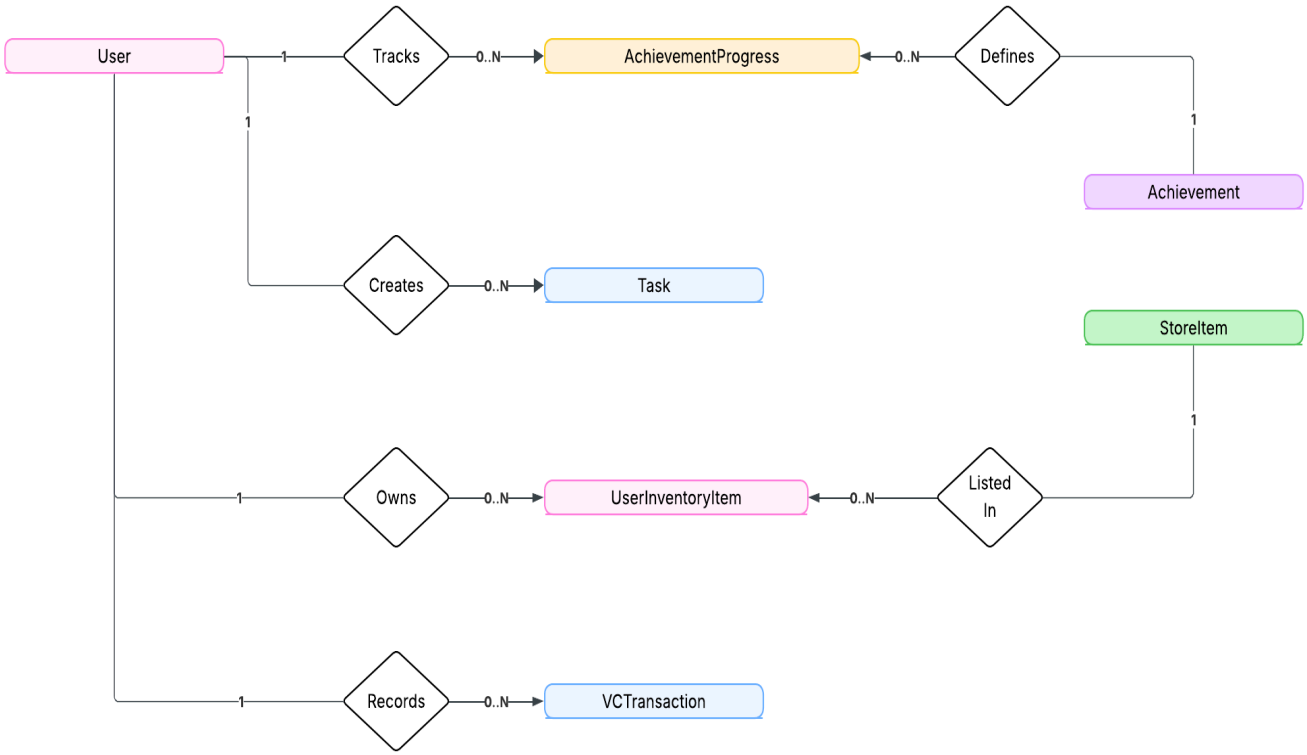
William Rice:

For more of a close-up and the actual link:
https://docs.google.com/document/d/1QC7zUvhwlNJ2829Kcl10NweNaj70lYhEJX7ejAQY8Gw/edit?usp=sharing

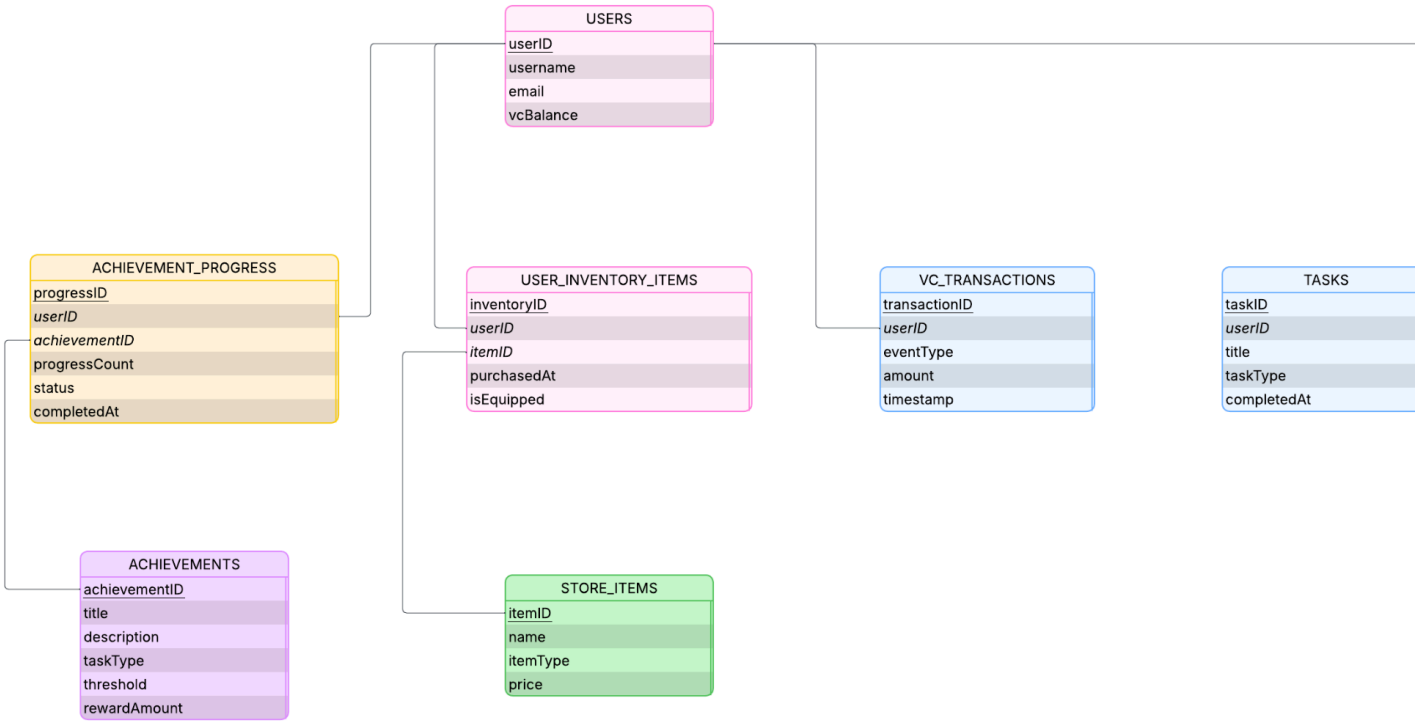
Achievement Tracking and Reward Class Diagram


User
+ userID: int
+ username: String
- email: String
+ vcBalance: int
+ tasks: List <Task>
+ progressList: List <AchievementProgress>
+ transactions: List <VCTransaction>
+ inventory: List <UserInventoryItem>
+ addVCBalance(amount: int): void
+ deductBalance(amount: int): void
+ hasSufficientBalance(price: int): boolean
composes 1 --> 0..*
AchievementProgress
- progressID: int
- progressCount: int
- status: String
- completedAt: DateTime
- achievement: Achievement
+ incrementProgress(): void
- checkThreshold(threshold: int): boolean
- markCompleted(): void
associated * --> 1
Achievement
- achievementID: int
- title: String
- description: String
- taskType: String
- threshold: int
- rewardAmount: int
+ matchesTaskType(taskType: String): boolean
composes 1 --> 0..*
UserInventoryItem
+ inventoryID: int
- purchasedAt: DateTime
- isEquipped: boolean
- item: StoreItem
+ equipItem(): void
+ unequipItem(): void
associated * --> 1
StoreItem
- itemID: int
- name: String
- itemType: String
- price: int
+ isAvailable(): boolean
composes 1 --> 0..*
Task
- taskID: int
- title: String
- taskType: String
- completedAt: DateTime
+ complete(): void
VCTransaction
- transactionID: int
- eventType: String
- amount: int
- timestamp: DateTime
+ logTransaction(): void
associated 1 --> 0..*

ER Diagram

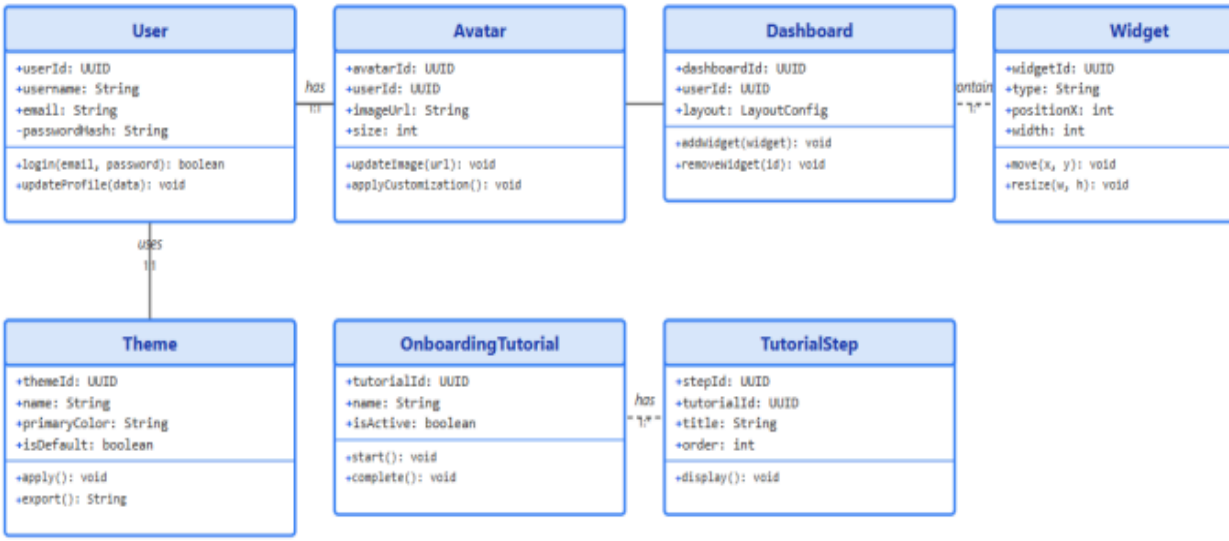

User
1
Tracks
0..N
AchievementProgress
0..N
Defines
1
Achievement
1
Creates
0..N
Task
StoreItem
1
1
Owns
0..N
UserInventoryItem
0..N
Listed In
1
Records
0..N
VCTransaction

Database Tables

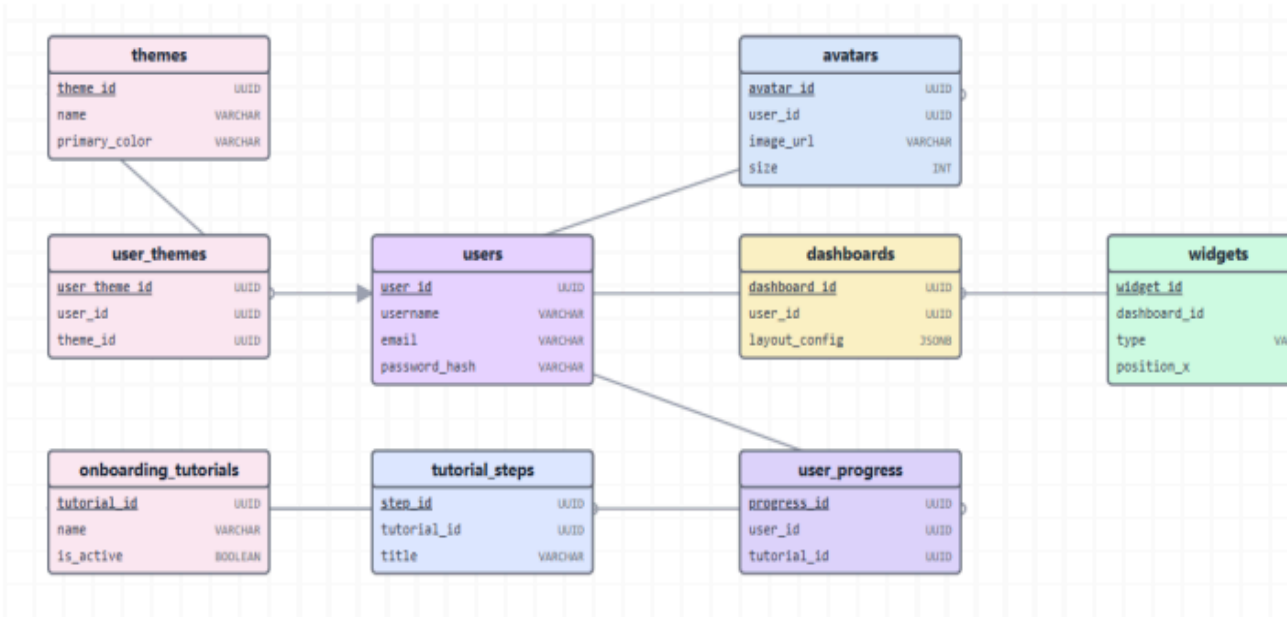


Raj Tatavarthi:

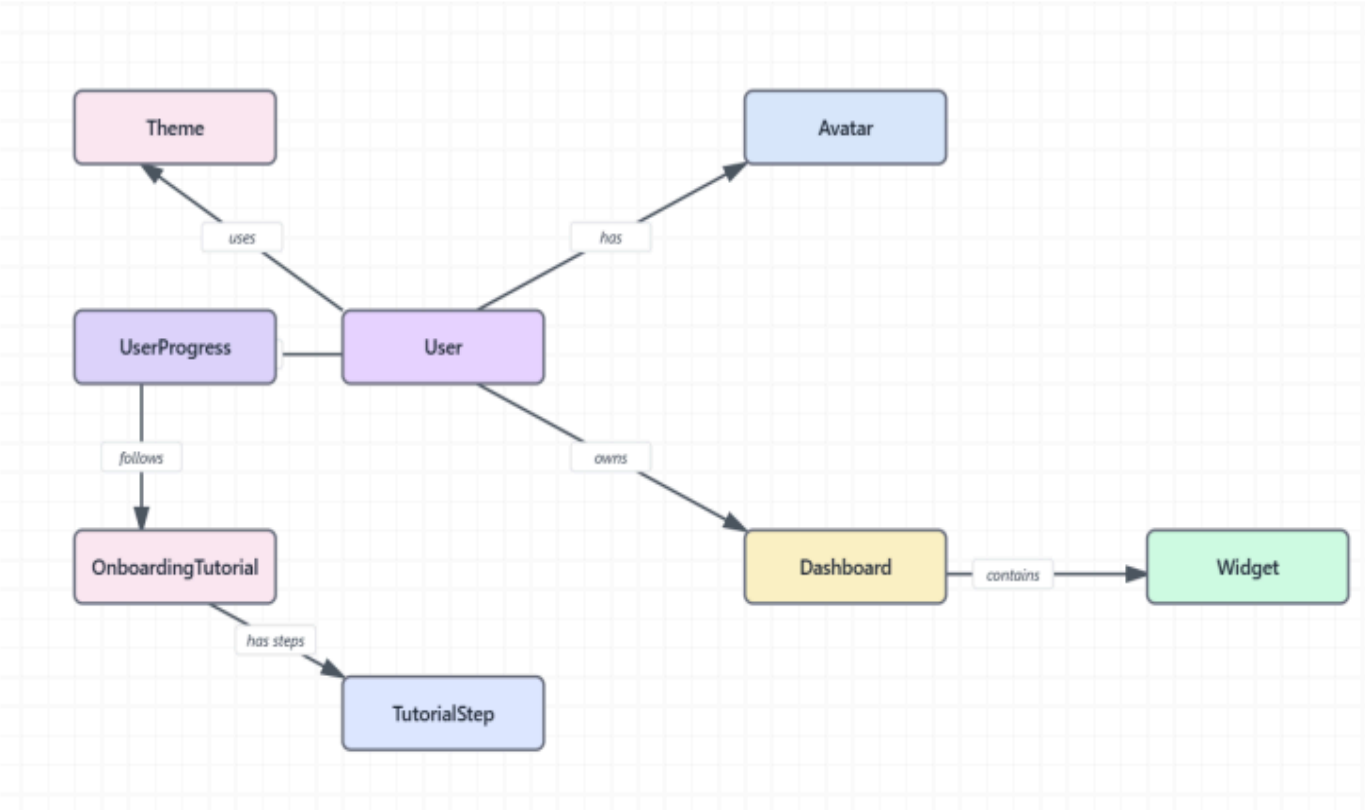
Class digram



Database table

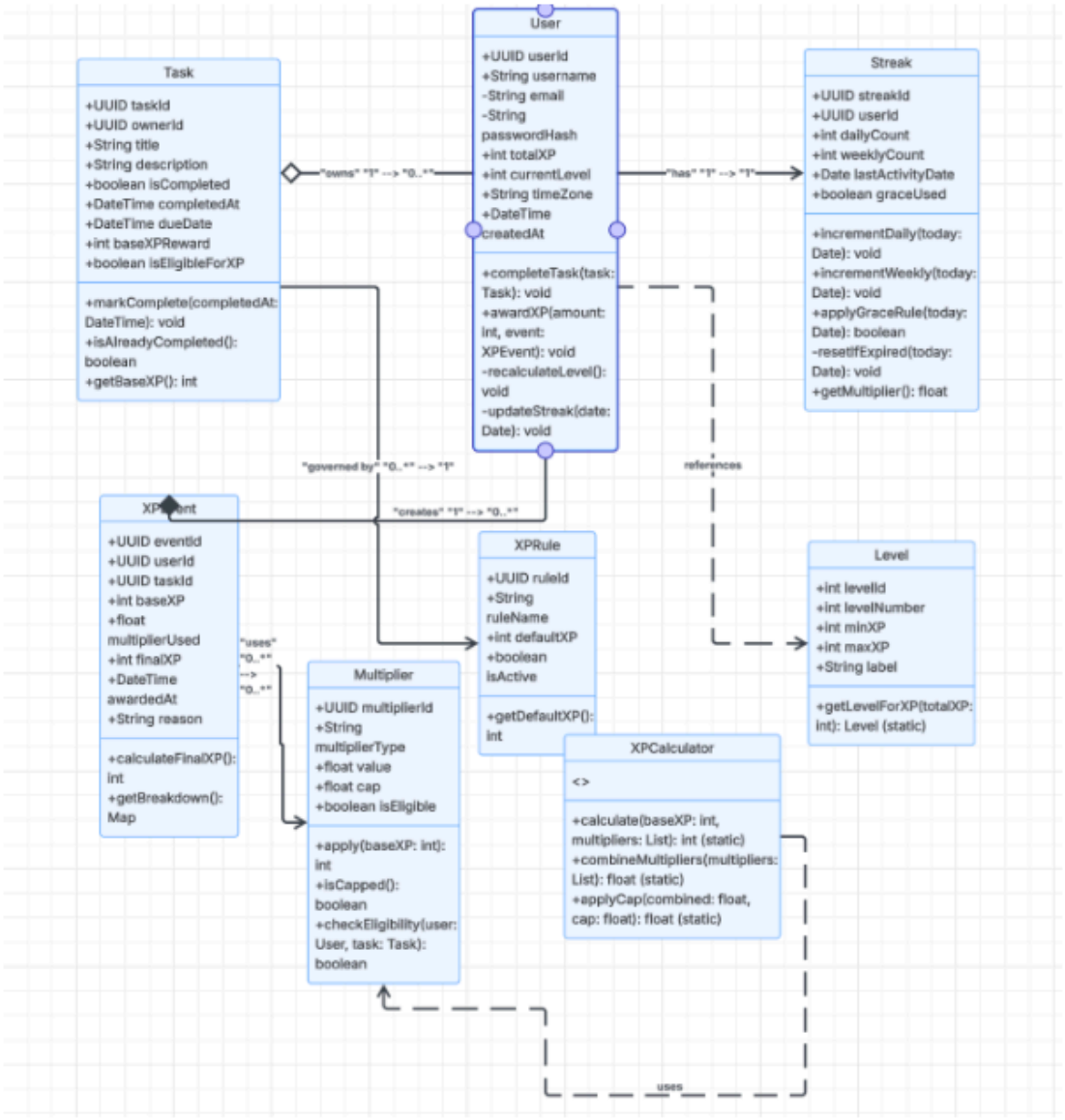


ER Diagram:

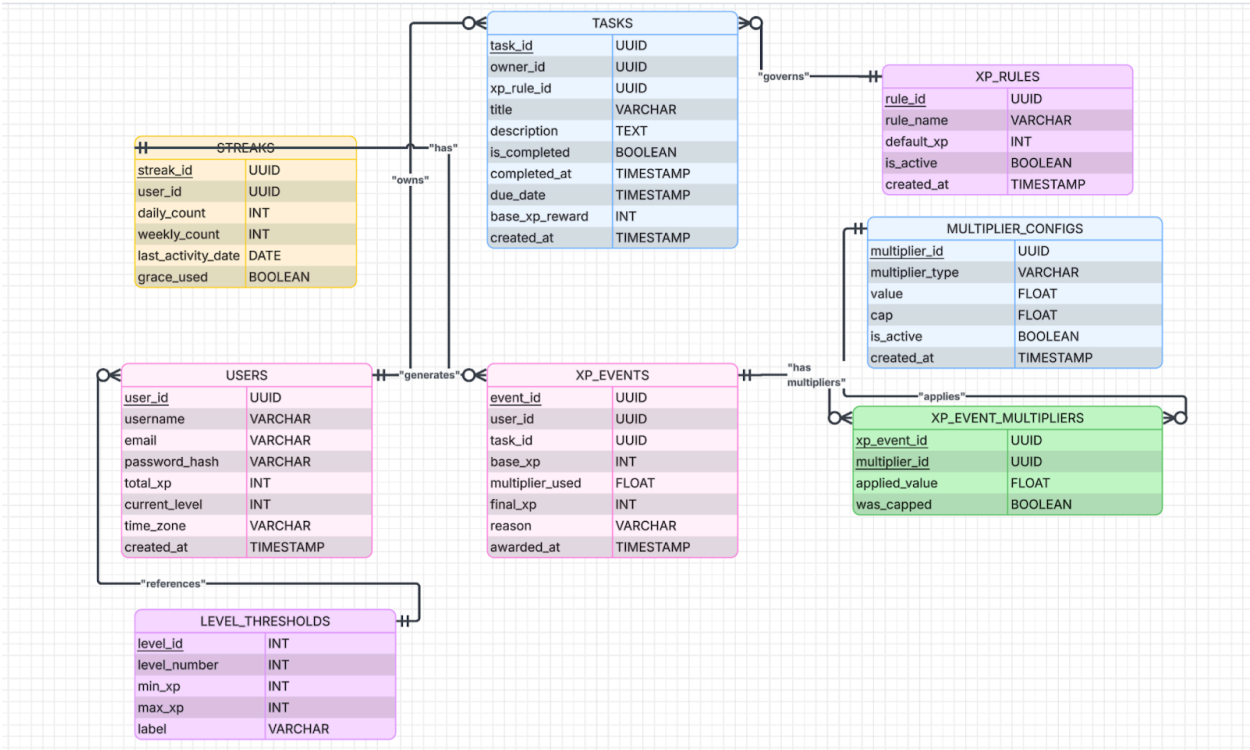

Theme
Avatar
uses
has
UserProgress
User
follows
owns
OnboardingTutorial
Dashboard
contains
Widget
has steps
TutorialStep

Vivan Verma:

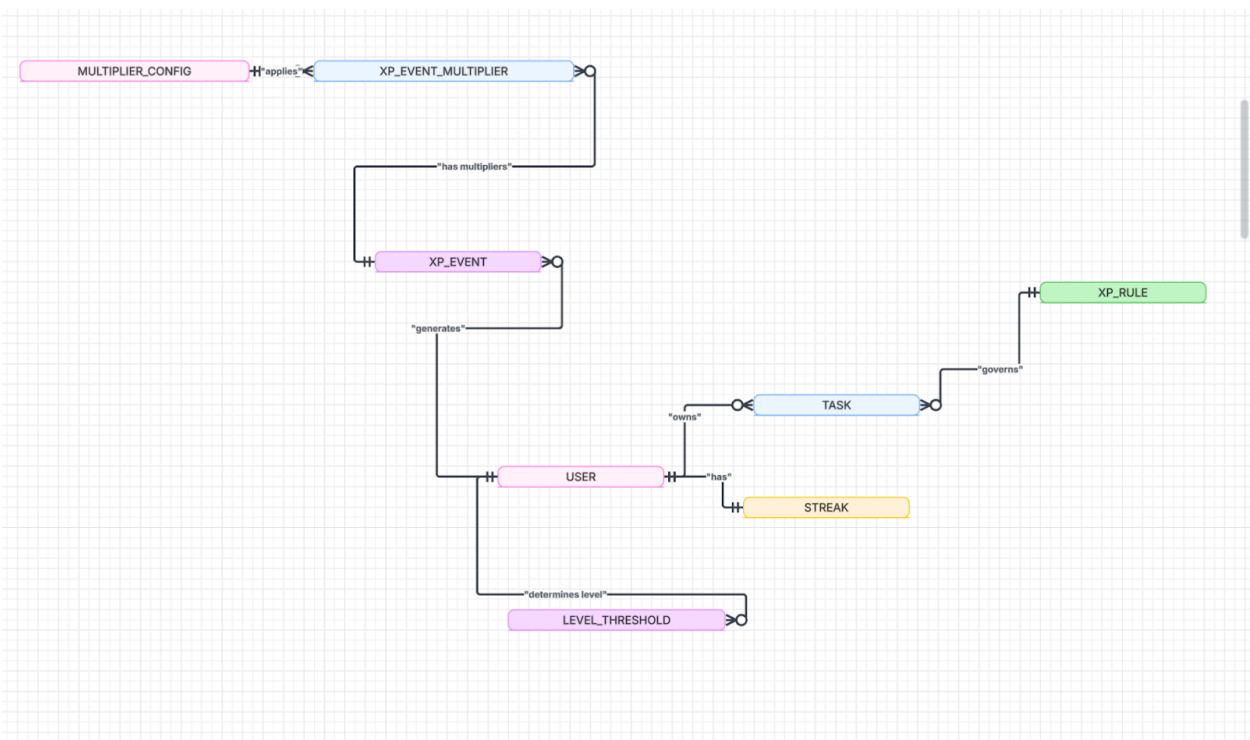
Class Diagram



Database Tables


TASKS
task_id UUID
owner_id UUID
xp_rule_id UUID
title VARCHAR
description TEXT
is_completed BOOLEAN
completed_at TIMESTAMP
due_date TIMESTAMP
base_xp_reward INT
created_at TIMESTAMP
"governs"
XP_RULES
rule_id UUID
rule_name VARCHAR
default_xp INT
is_active BOOLEAN
created_at TIMESTAMP
STREAKS
streak_id UUID
user_id UUID
daily_count INT
weekly_count INT
last_activity_date DATE
grace_used BOOLEAN
"has"
"owns"
MULTIPLIER_CONFIGS
multiplier_id UUID
multiplier_type VARCHAR
value FLOAT
cap FLOAT
is_active BOOLEAN
created_at TIMESTAMP
USERS
user_id UUID
username VARCHAR
email VARCHAR
password_hash VARCHAR
total_xp INT
current_level INT
time_zone VARCHAR
created_at TIMESTAMP
"generates"
XP_EVENTS
event_id UUID
user_id UUID
task_id UUID
base_xp INT
multiplier_used FLOAT
final_xp INT
reason VARCHAR
awarded_at TIMESTAMP
"has multipliers"
"applies"
XP_EVENT_MULTIPLIERS
xp_event_id UUID
multiplier_id UUID
applied_value FLOAT
was_capped BOOLEAN
"references"
LEVEL_THRESHOLDS
level_id INT
level_number INT
min_xp INT
max_xp INT
label VARCHAR

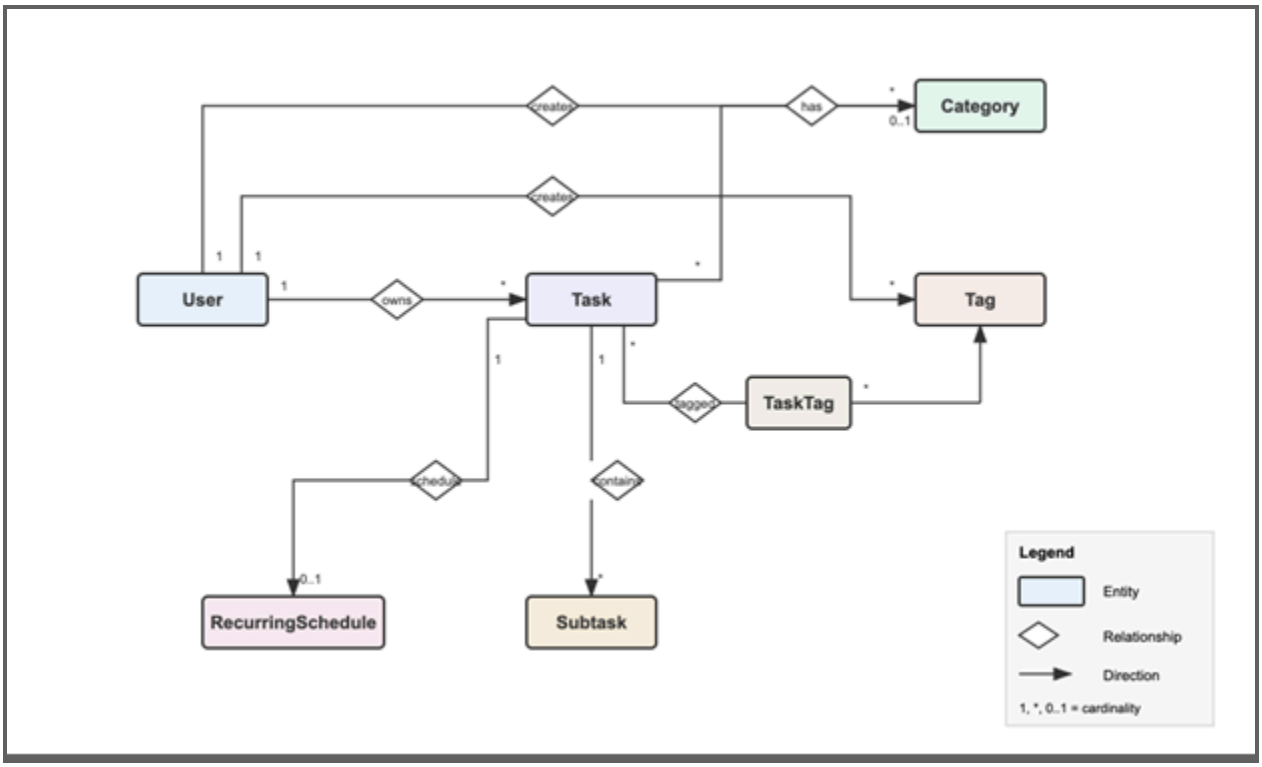
ER Diagram


MULTIPLIER_CONFIG
"applies"
XP_EVENT_MULTIPLIER
"has multipliers"
XP_EVENT
"generates"
XP_RULE
"governs"
TASK
"owns"
USER
"has"
STREAK
"determines level"
LEVEL_THRESHOLD

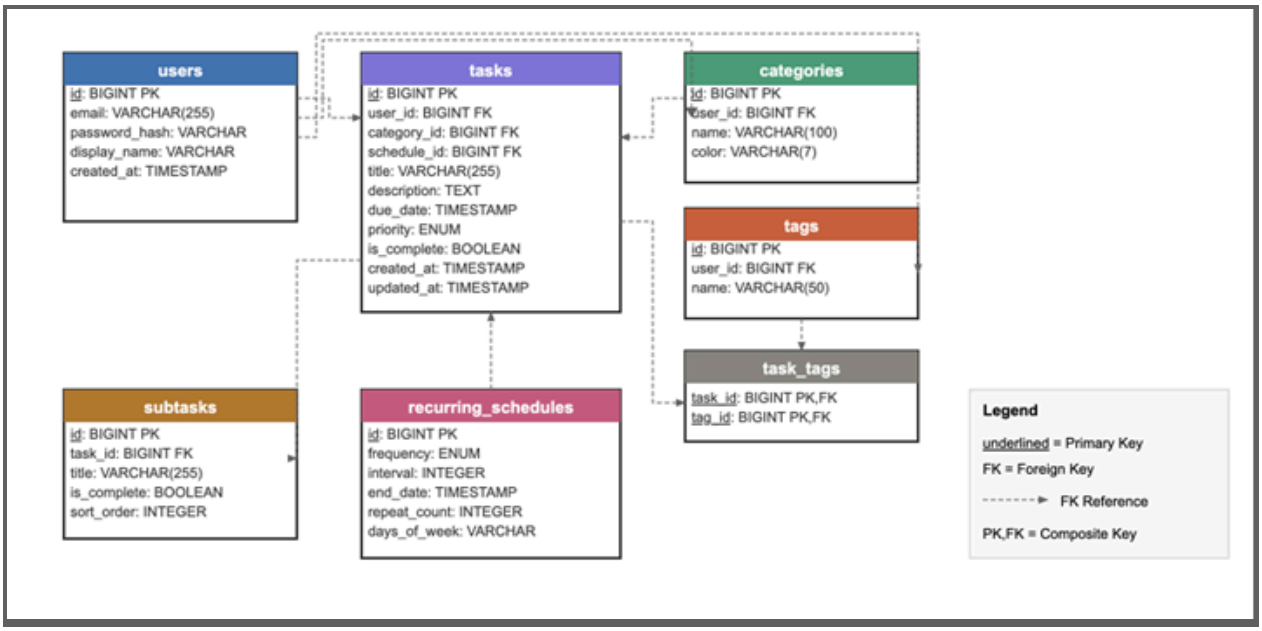
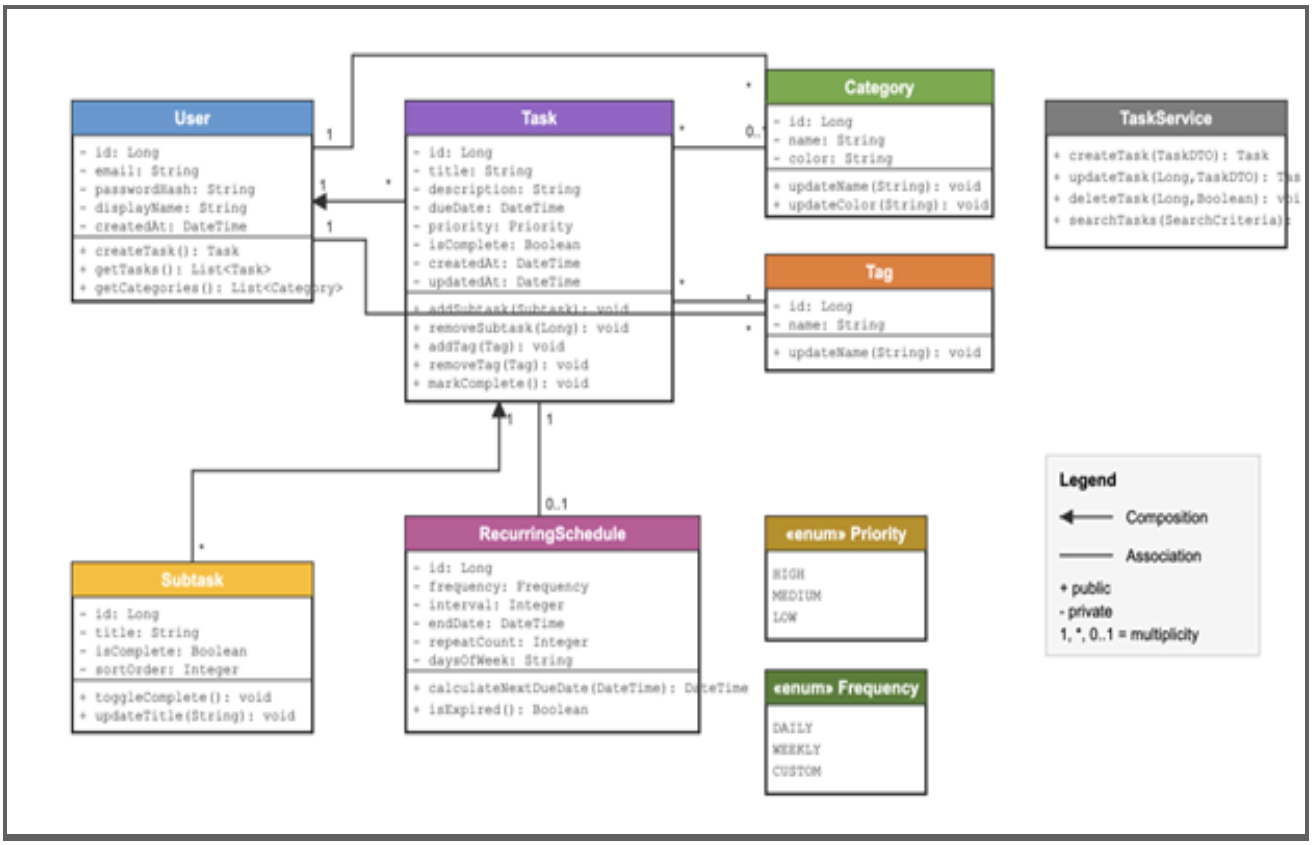
Class names changed, MultiplierRule became just Multiplier and XPRule was added as a separate class. Fields were significantly expanded, the original has minimal fields so I added more. New classes were added, XPCalculator, Streak as a fuller class and Level with full threshold fields. XPEvent fields were expanded. Overall I did my best to improve the diagram by making it denser and more realistic.

Rochan Sonti:

Task Management System - ER Diagram



Task Management System - Database Tables



Task Management System - Class Diagram